Daedalus | Structures Lead

OBJECTIVE

Stable, recoverable rocket
4,000-ft design target

MY OWNERSHIP

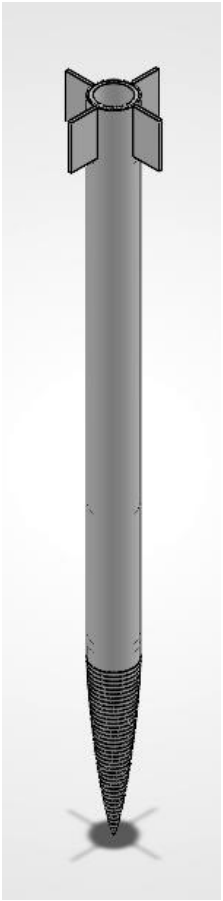
- Von Kármán nose + shoulder
- Fin can + structural integration
- Recovery attachment + interfaces
- PLA build, fit testing + validation

ENGINEERING CHALLENGES

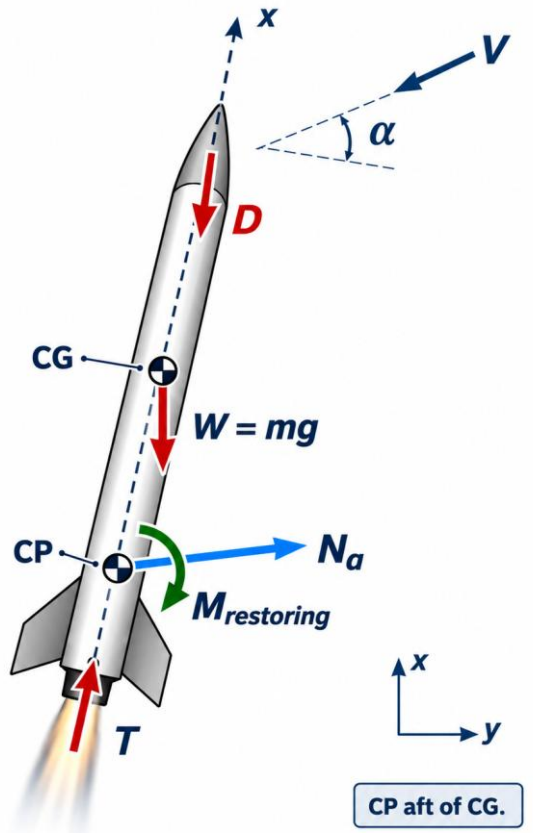
- Nose: drag vs. mass and printability
- Fin alignment after design review
- Simulation assumptions vs. testing

4,086 ft simulated **1.23 cal** modeled

SOLIDWORKS CAD

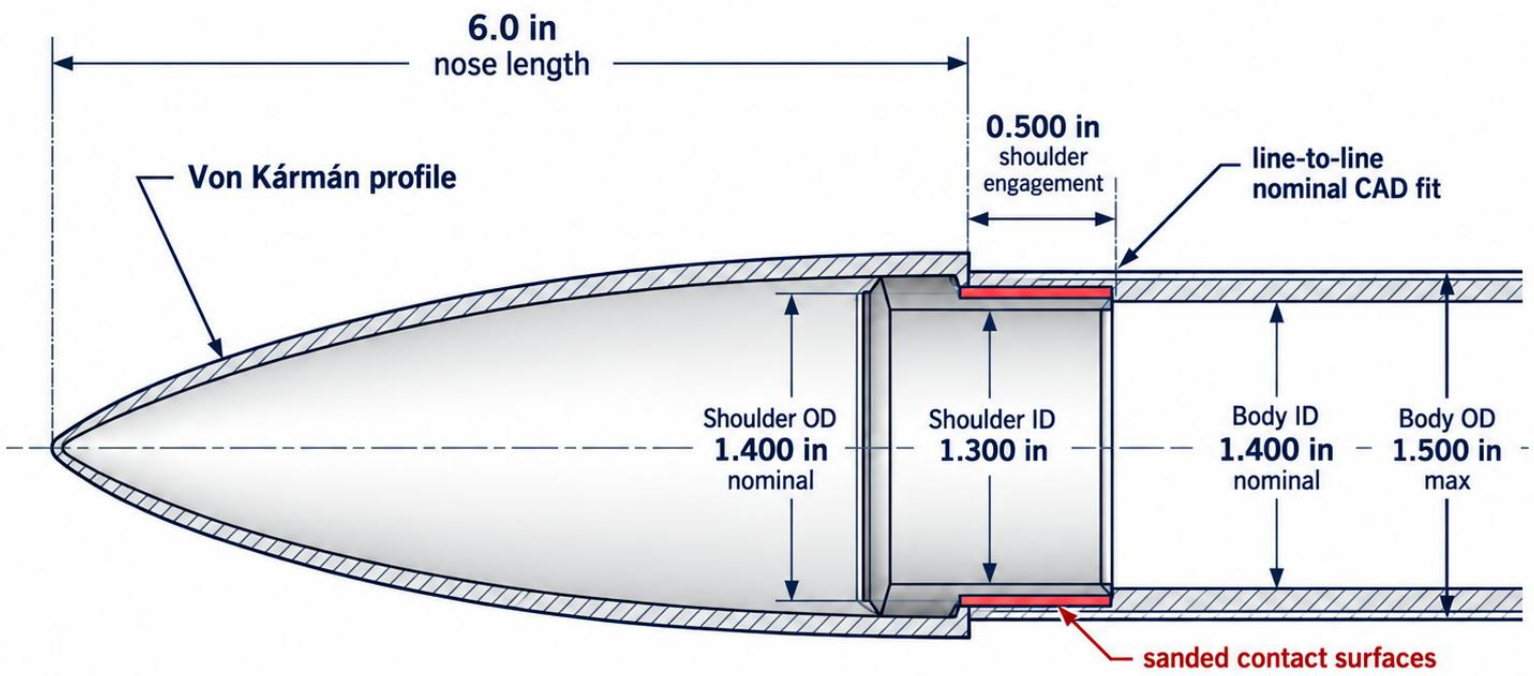


FORCES + STABILITY

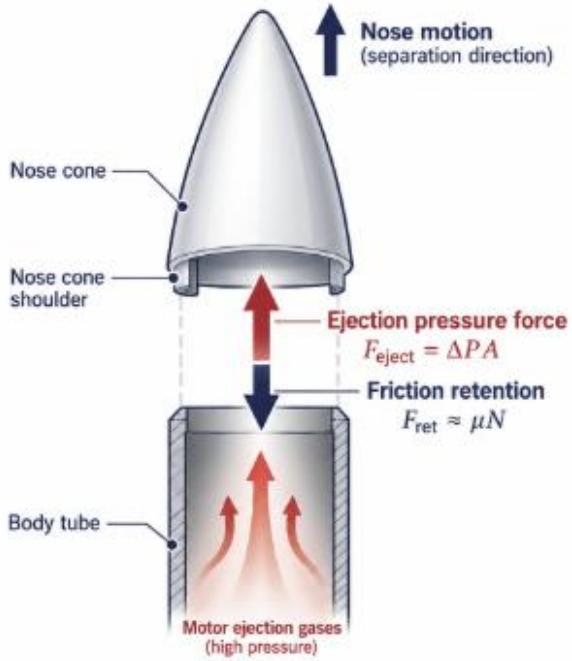


Built + ground-tested • never flown

Nose-Cone Design & Retention Testing



NOSE SEPARATION FBD



PLA print → sand → spring scale

≈19 N extraction

recalled; original readings unavailable

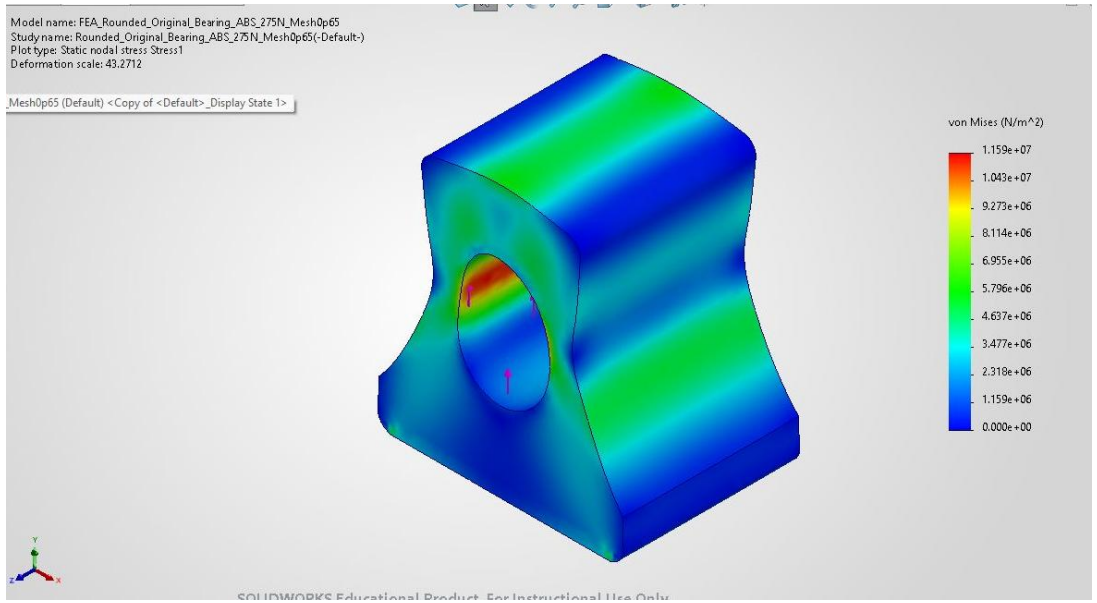
$$F_{ret} \approx \mu N$$

$$\Delta P_{min} = \frac{F_{pull}}{A} = \frac{19}{9.931 \times 10^{-4}} = 19.1 \text{ kPa} = 2.77 \text{ psi}$$

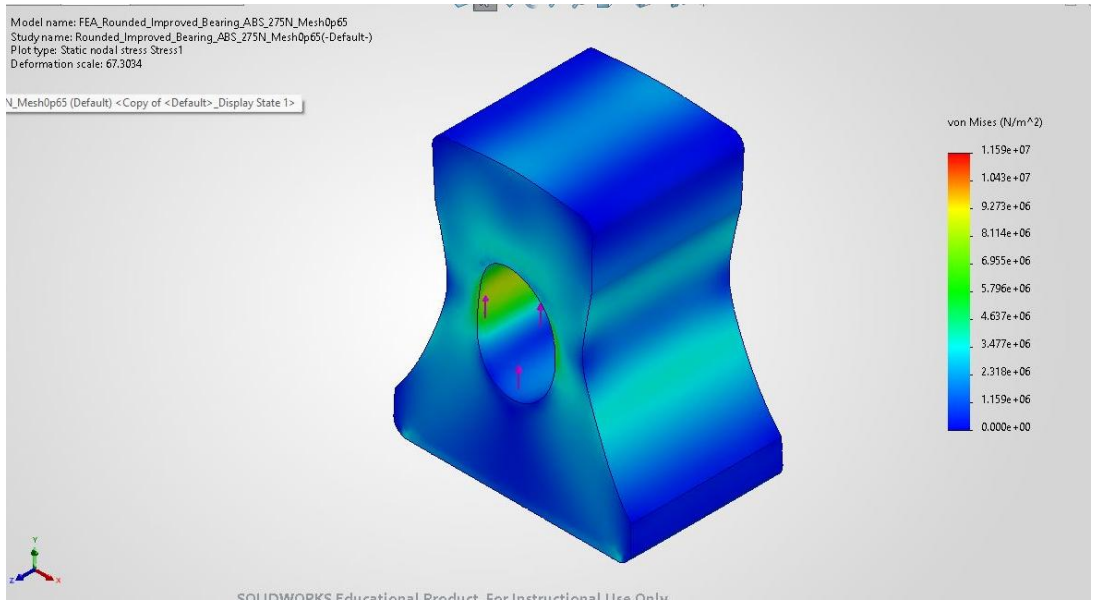
Motor delayed ejection • pressure and gas path unverified

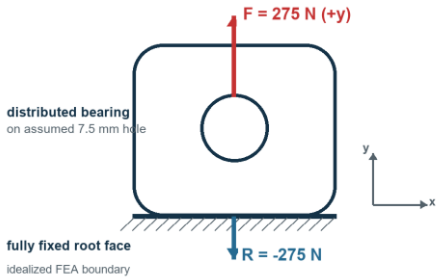
Recovery Eyelet Design & Structural Analysis

ORIGINAL-STYLE



WIDER PROFILE + HOLE LIGAMENT





−23.5% **11.591 → 8.869 MPa peak**

−20.3% **0.04138 → 0.03299 mm**

NEXT VALIDATION

- CFD + fin optimization
- Repeat fit + deployment tests
- Printed PLA + full attachment

Assumed 275 N • fixed root • isotropic ABS | Wider eyelet not manufactured